

**HUSSEINI SULEIMAN**

**2025/FWSD/11478**

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**DATE: 4<sup>TH</sup> September, 2026**

**COURSE CODE: SBT-DF202 · PRACTICAL LABORATORIES**

Lab 3 — Data Carving with XXD, Binwalk and Scalpel

Submit before 05 Sep 2026, 4:20 PM WAT.

## **SCENARIO**

### **Your practical context**

A forensic storage image contains files that may no longer be recoverable through normal file-system browsing. Some file metadata may be missing, damaged, or unreliable. Acting as a junior digital forensic analyst, you are required to examine raw file content, identify file signatures, recover embedded content, and carve deleted files from a USB forensic image while documenting every step in a repeatable forensic manner.

## **OBJECTIVES**

### **What you must demonstrate**

By completing this lab, students will demonstrate their ability to explain why data carving is required when file-system metadata is missing or unreliable; use xxd to examine binary file content and identify headers and footers; recognise JPEG SOI and EOI signatures; reconstruct binary content from a hexadecimal dump; use The Sleuth Kit to examine inode, block and sector-level data; identify and extract embedded content using Binwalk; recover deleted files from a forensic USB image using Scalpel; verify recovered evidence using cryptographic hashes; and prepare a professional forensic report containing commands, screenshots, findings and limitations.

## INSTRUCTOR BRIEFING

### Laboratory guidance

Before beginning this lab, review the **Module 7 lessons and study material on Data Carving and File Recovery Techniques**. The lecture material is already a detailed walkthrough, so you are expected to understand file signatures, hexadecimal analysis, embedded-file recovery and carving concepts before executing commands.

Create a dedicated working directory and preserve all downloaded evidence files. Record MD5 and SHA-256 hashes before beginning analysis.

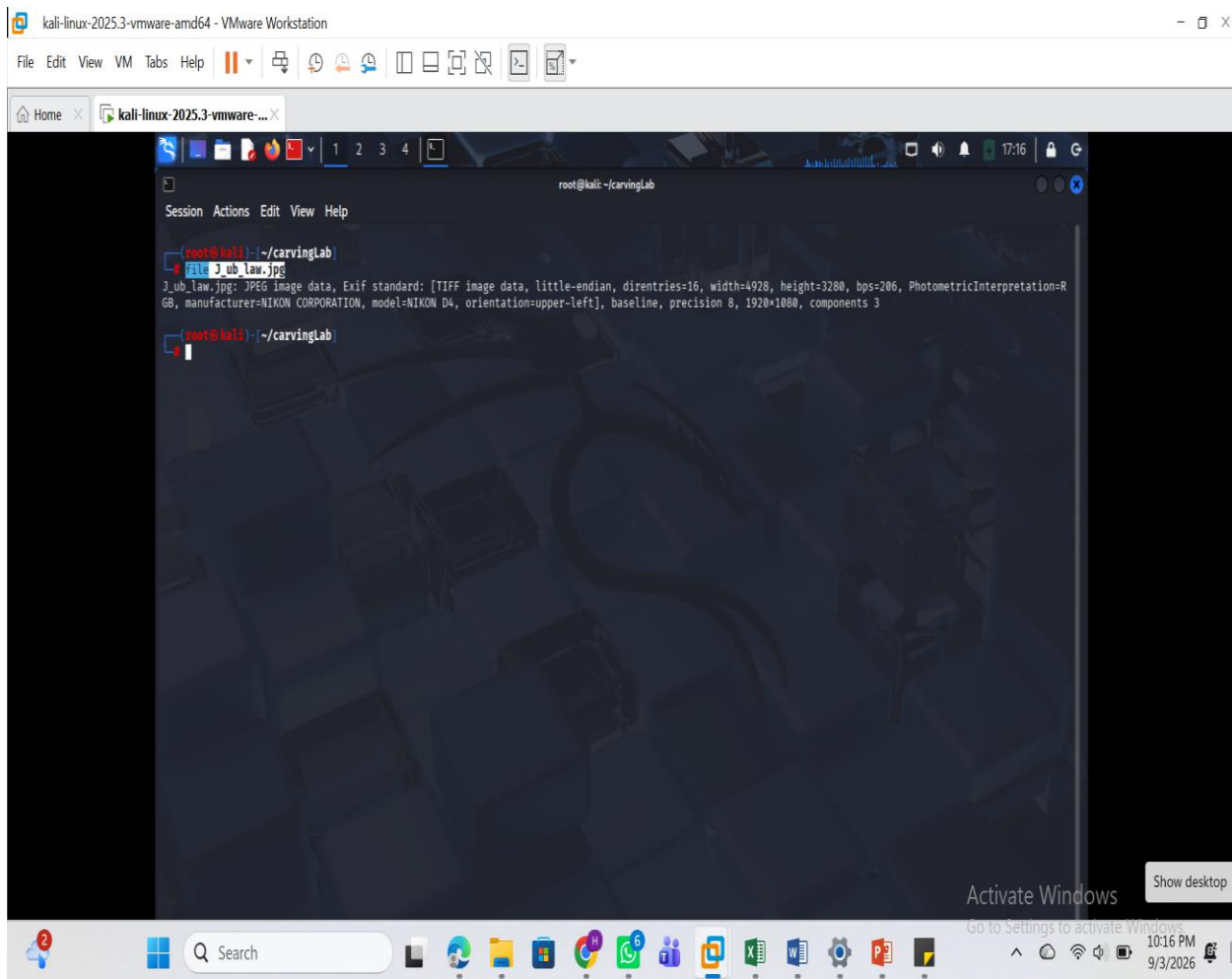
### Required Lab Resources

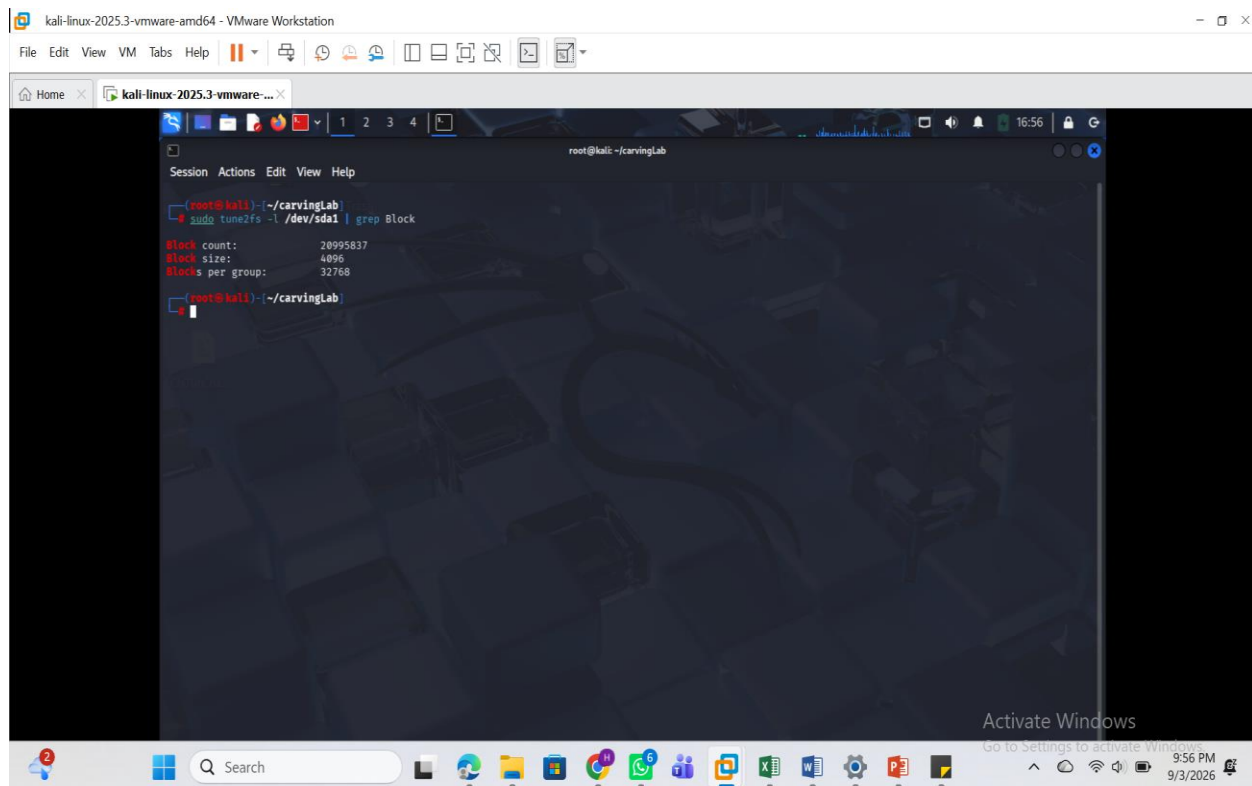
Download the authorised training evidence below:

- [Ch01InChap01.dd — Forensic DD Image](#)
- **J\_ub\_law.jpg — JPEG Sample for Hex and Signature Analysis**
- **120M.7z — USB Image for Scalpel File Carving**

These are the exact evidence resources identified in the lab guide.

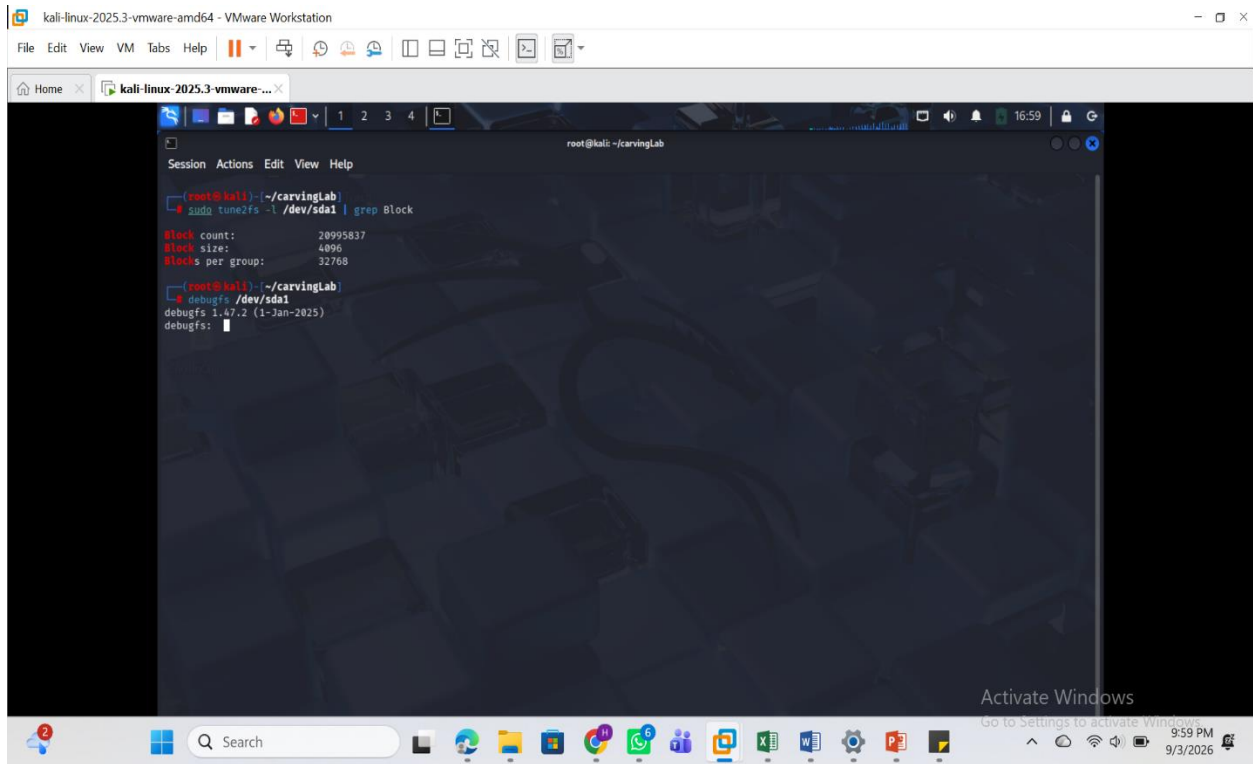
The display image when the command (J\_ub\_law.jpg) was inputted



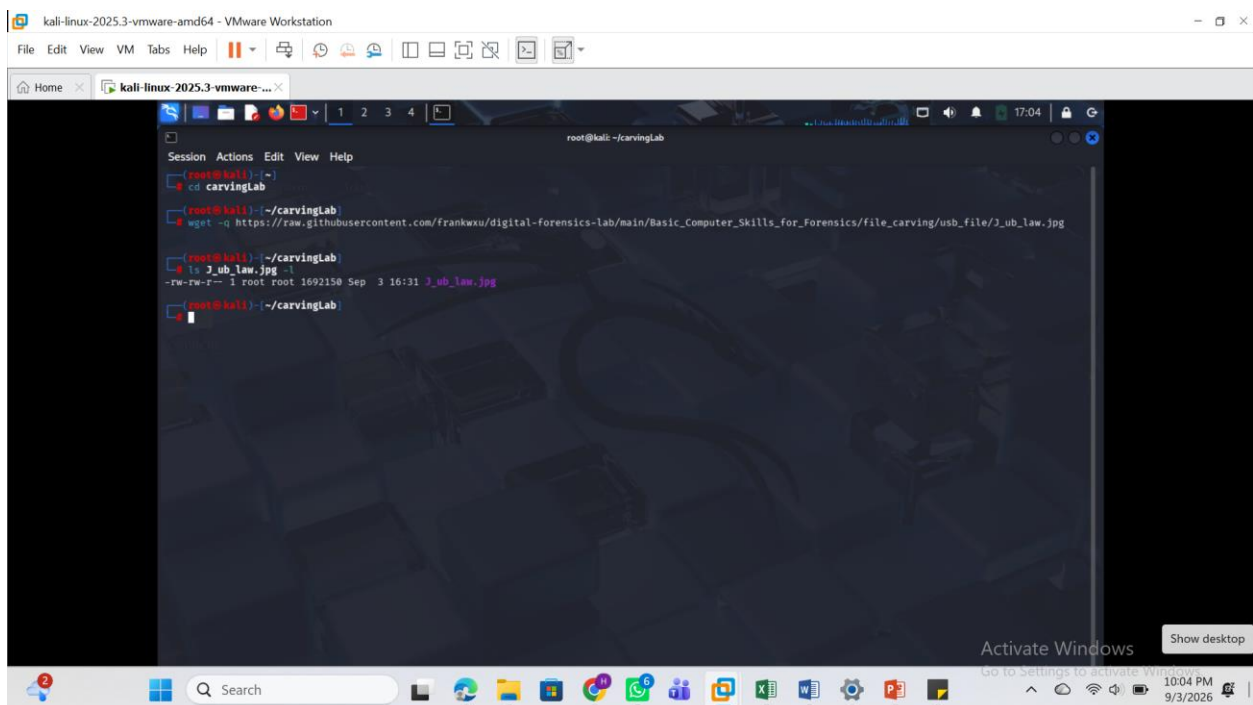


See the display image when the command is used: display J\_ub\_law.jpg

See the image info when the command (file J\_ub\_law.jpg) is used

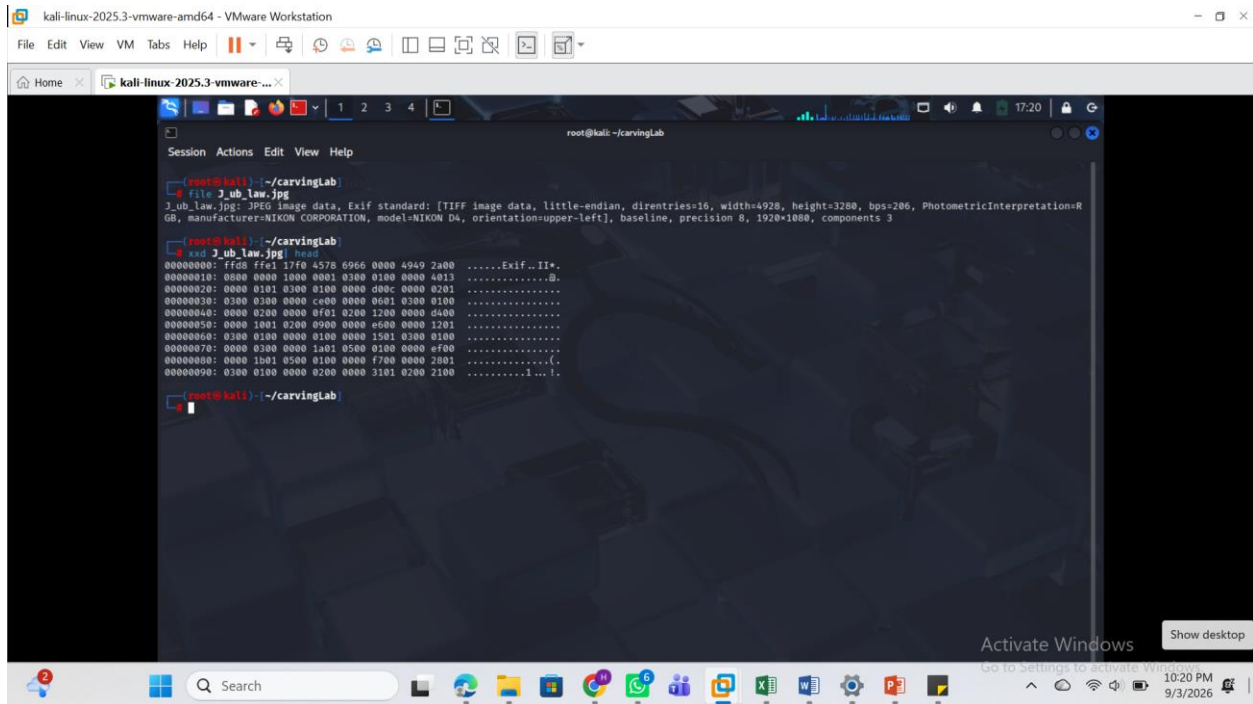


Using the command (`ls J_ub_law.jpg -l`) see the permissions and the lab environment preparation.



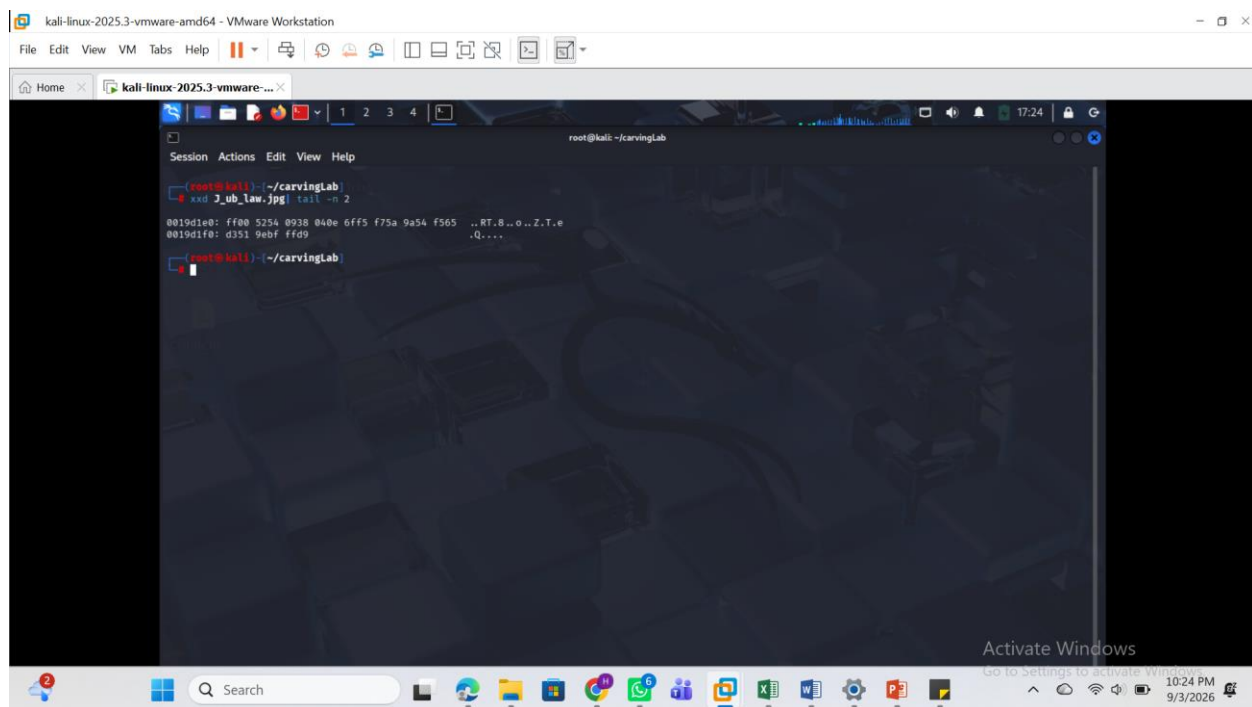
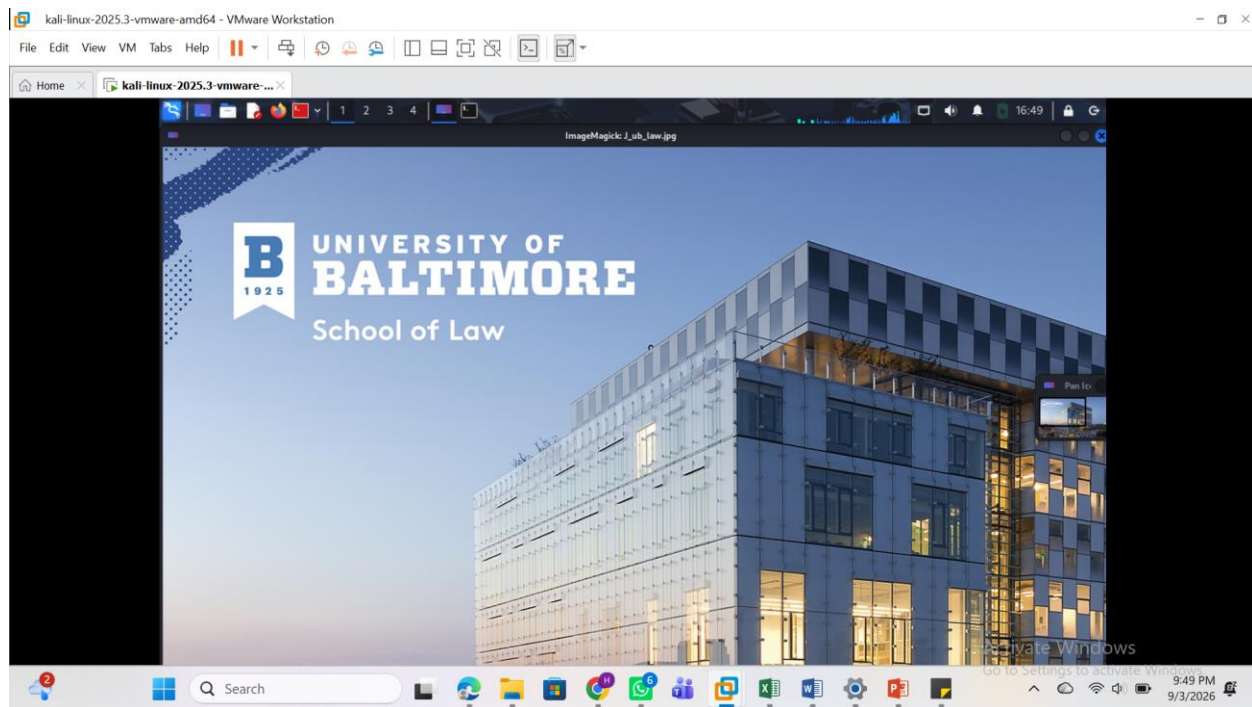
Using xxd to inspect J\_ub\_law.jpg, identify the JPEG header and footer signatures, create a plain hexadecimal dump, and reconstruct the image using reverse-hexdump functionality.

Using the command: `xxd J_ub_law.jpg | head`



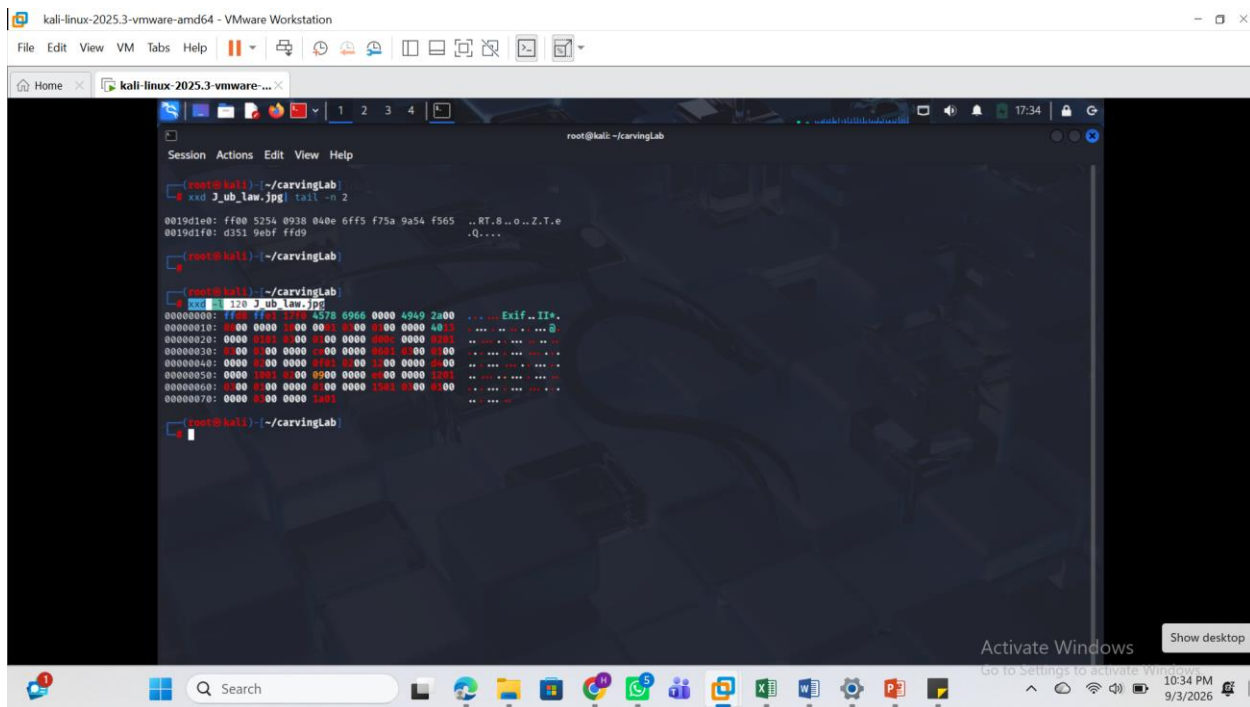
```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
root@kali: ~/carvingLab
root@kali:~/carvingLab# file J_ub_law.jpg
J_ub_law.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=280, PhotometricInterpretation=R
GB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3
root@kali:~/carvingLab# xxd J_ub_law.jpg | head
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ....Exif..II*.
00000010: 0000 0000 1000 0001 0300 0100 0000 4013  .........B.
00000020: 0000 0101 0300 0100 0000 000c 0000 0201  ....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100  ....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400  ....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201  ....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100  ....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00  ....
00000080: 0000 1501 0500 0100 0000 f700 0000 2601  ....(.
00000090: 0300 0100 0000 0200 0000 3101 0200 2100  ....1...).
```

Command to show the tail of the picture: `xxd J_ub_law.jpg | tail -n 2`

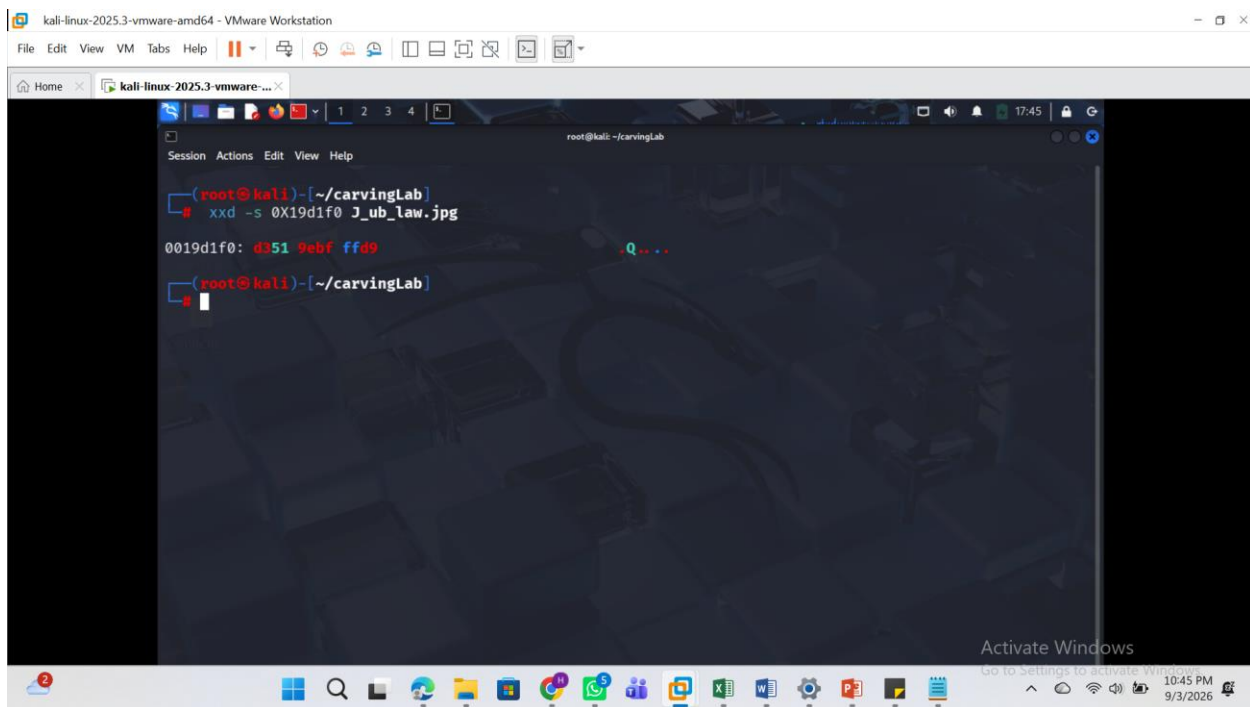


Using the command: `xxd -l 120 J_ub_law.jpg`





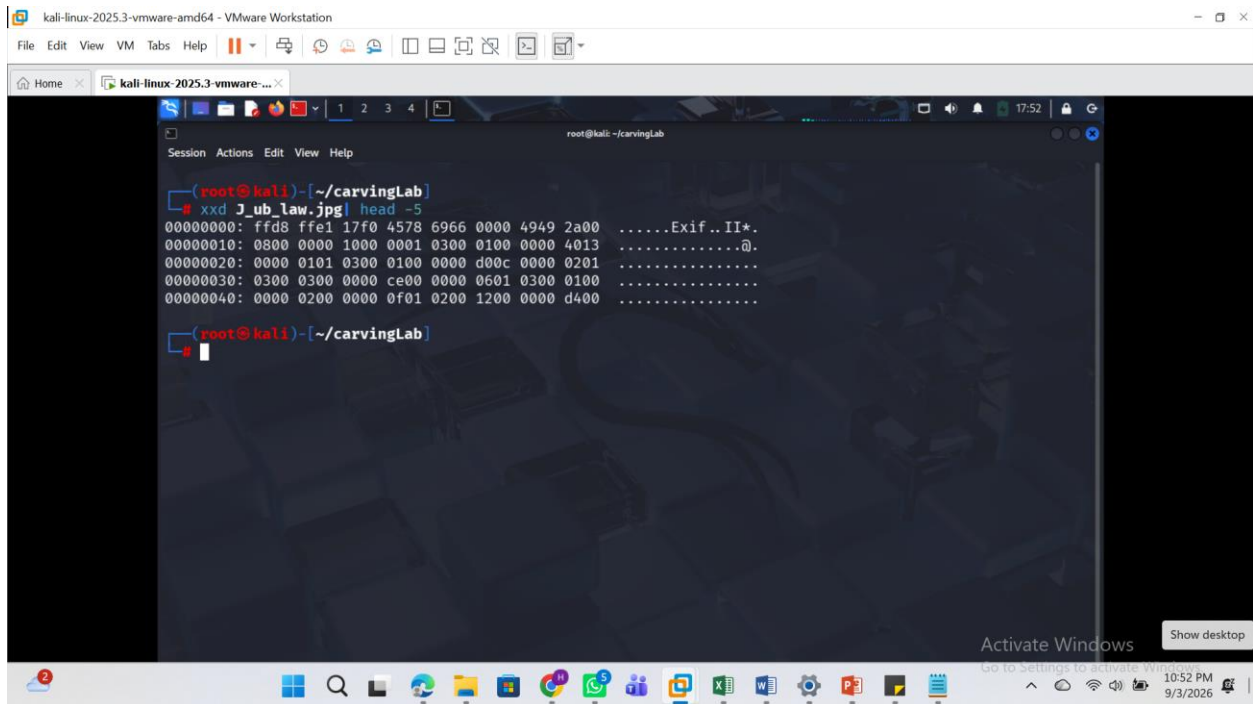
Using the command: `xxd -s 0x19d1f0 J_ub_law.jpg`





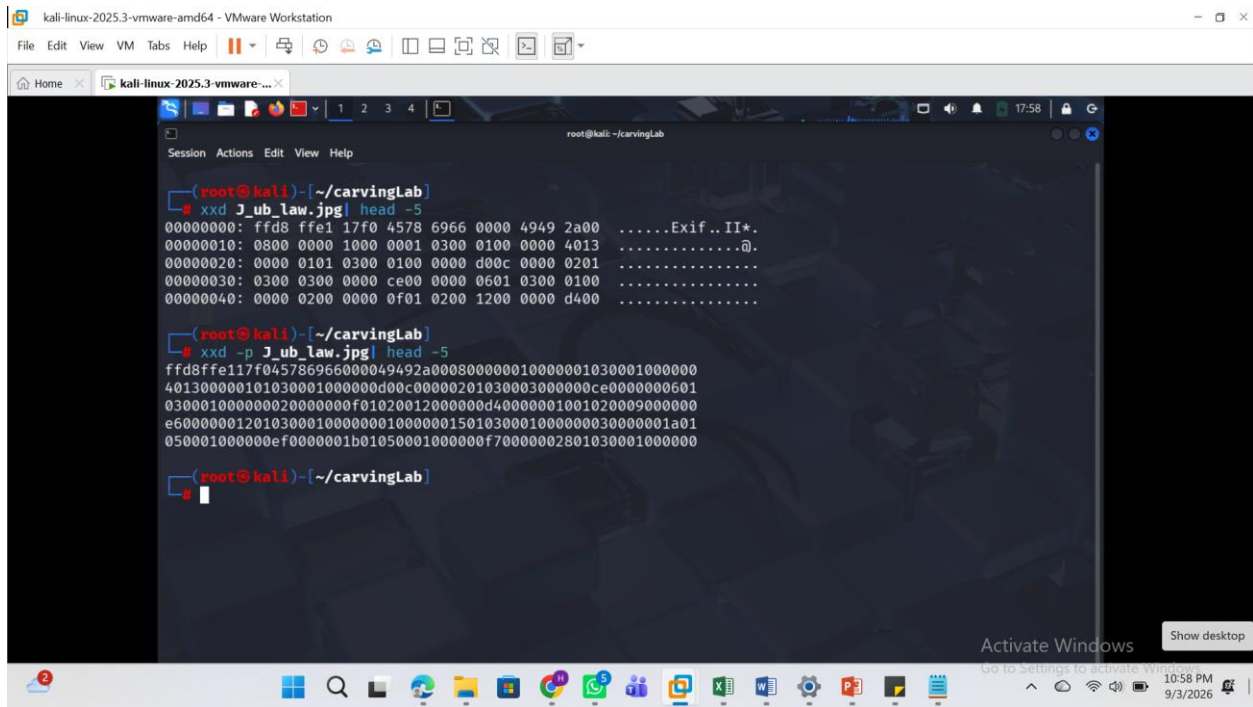
## Using the commands to Search signatures in plain hexdump

: xxd J\_ub\_law.jpg | head -5



```
root@kali: ~/carvingLab
root@kali)~/.carvingLab
# xxd J_ub_law.jpg | head -5
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ....Exif..II*
00000010: 0800 0000 1000 0001 0300 0100 0000 4013  .....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201  .....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100  .....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400  .....

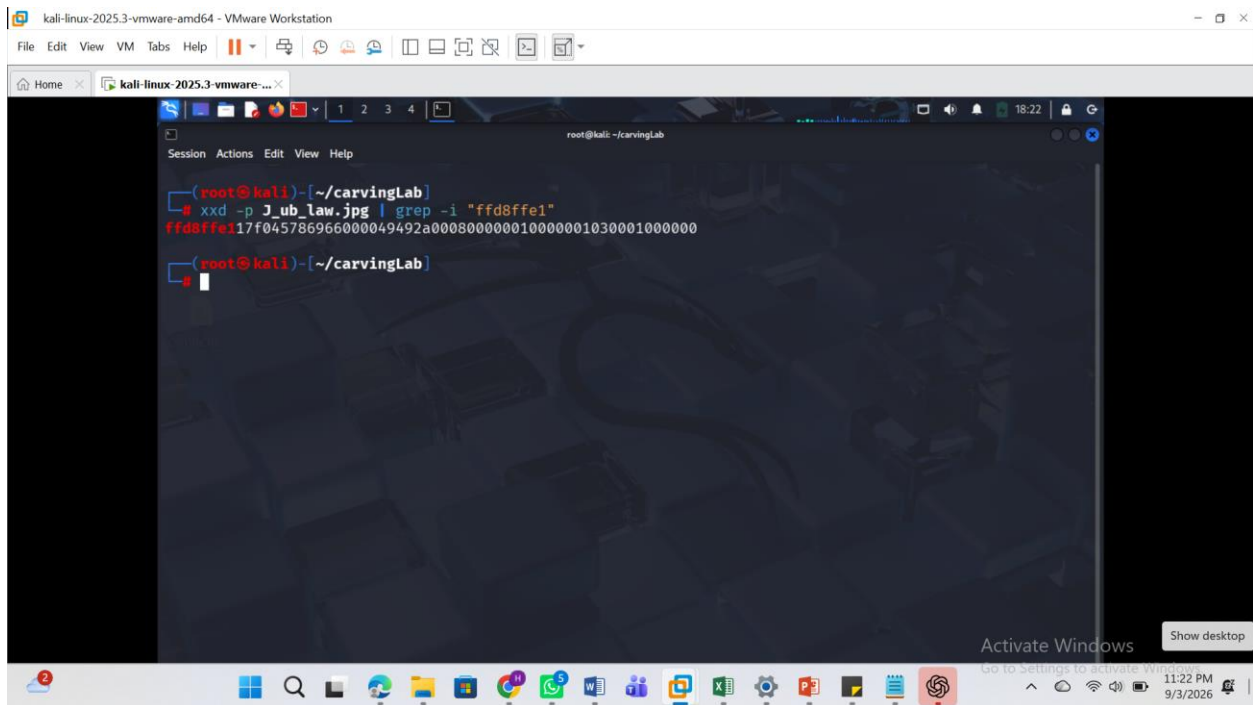
root@kali)~/.carvingLab
#
```



```
root@kali:~/carvingLab
# xxd J_ub_law.jpg | head -5
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 .....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201 .....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100 .....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400 .....

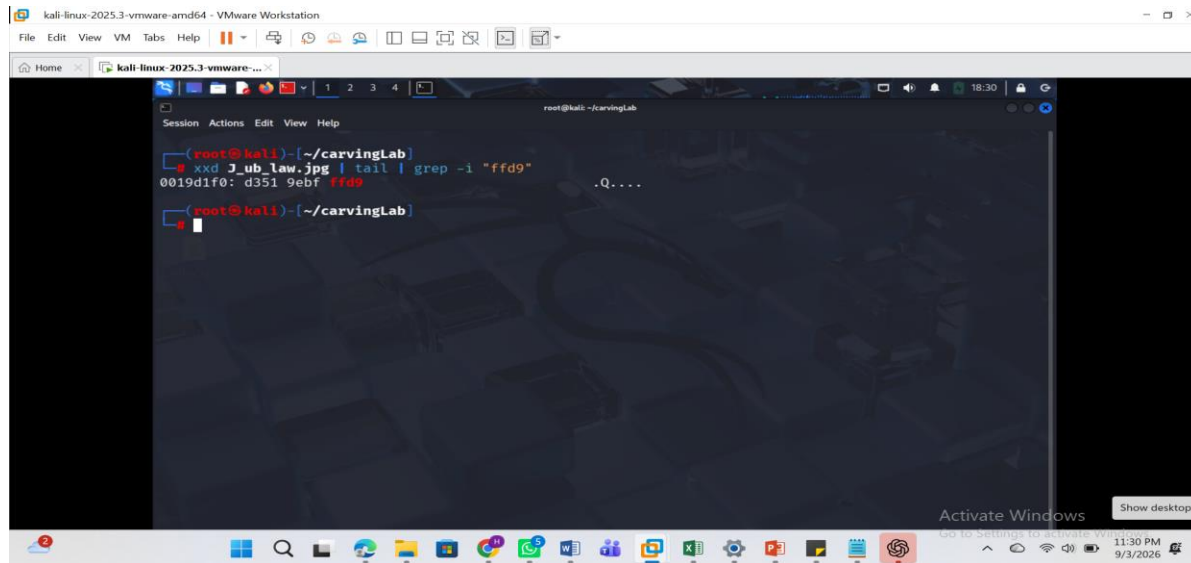
root@kali:~/carvingLab
# xxd -p J_ub_law.jpg | head -5
ffd8ffe117f045786966000049492a000800000010000001030001000000
401300000101030001000000d00c00000201030003000000ce0000000601
03000100000002000000f01020012000000d40000001001020009000000
e6000000120103000100000001000000150103000100000030000001a01
050001000000ef0000001b01050001000000f70000002801030001000000
```

Search header in plain hexdump using the command: `xxd -p J_ub_law.jpg | grep -i "ffd8ffe1"`



```
root@kali:~/carvingLab
# xxd -p J_ub_law.jpg | grep -i "ffd8ffe1"
ffd8ffe117f045786966000049492a000800000010000001030001000000
```

`xxd J_ub_law.jpg | tail | grep -i "ffd9"`

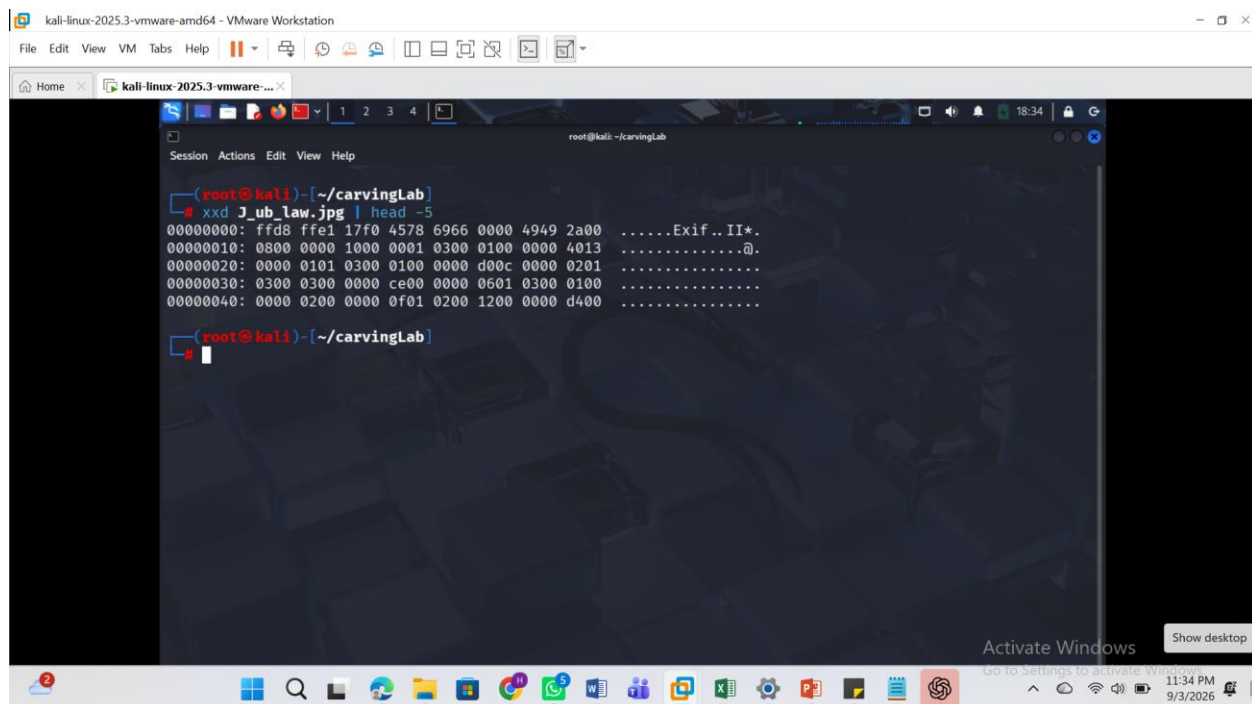


The screenshot shows a terminal window titled 'kali-linux-2025.3-vmware-amd64 - VMware Workstation'. The user is at the root prompt in the directory ~/carvingLab. The command `xxd J_ub_law.jpg | tail | grep -i "ffd9"` has been executed, resulting in the output `0019d1f0: d351 9ebf ffd9 .Q....`. The terminal window includes a menu bar (Session, Actions, Edit, View, Help) and a taskbar at the bottom with various application icons. An 'Activate Windows' watermark is visible in the bottom right corner.

```
root@kali: ~/carvingLab
# xxd J_ub_law.jpg | tail | grep -i "ffd9"
0019d1f0: d351 9ebf ffd9 .Q....

root@kali: ~/carvingLab
#
```

Using the command: `xxd J_ub_law.jpg | head -5`

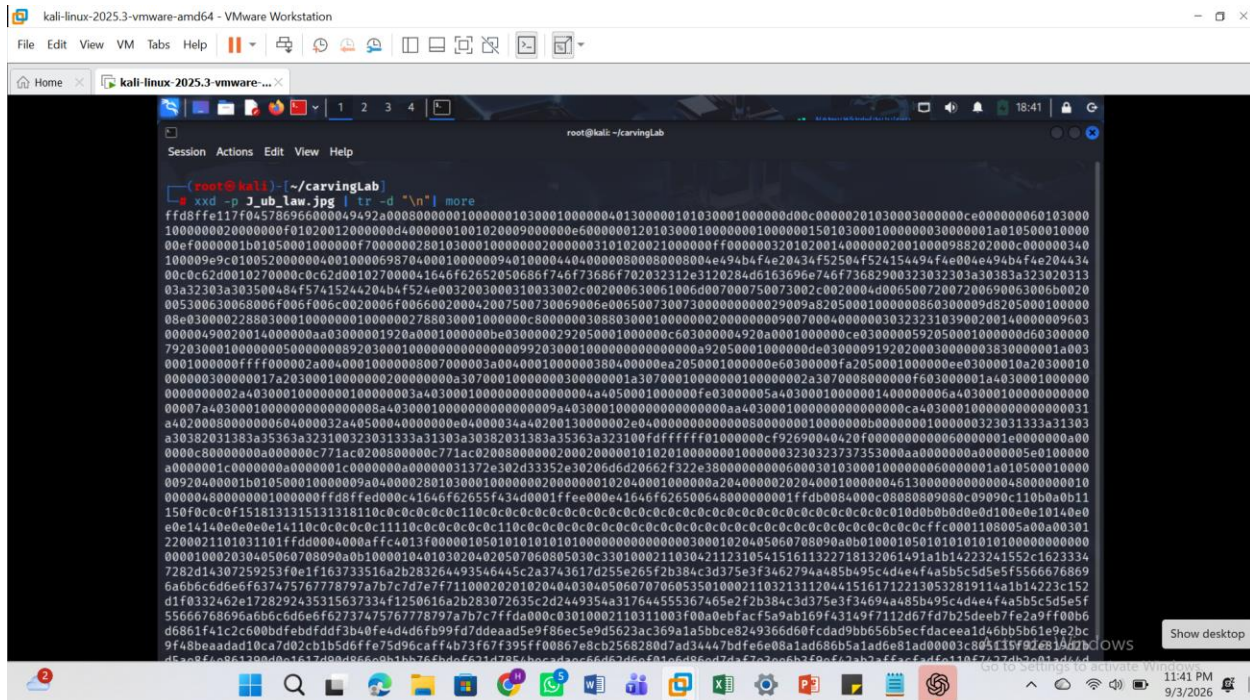


The screenshot shows a terminal window titled 'kali-linux-2025.3-vmware-amd64 - VMware Workstation'. The user is at the root prompt in the directory ~/carvingLab. The command `xxd J_ub_law.jpg | head -5` has been executed, resulting in the following output:

```
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 .....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201 .....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100 .....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400 .....
```

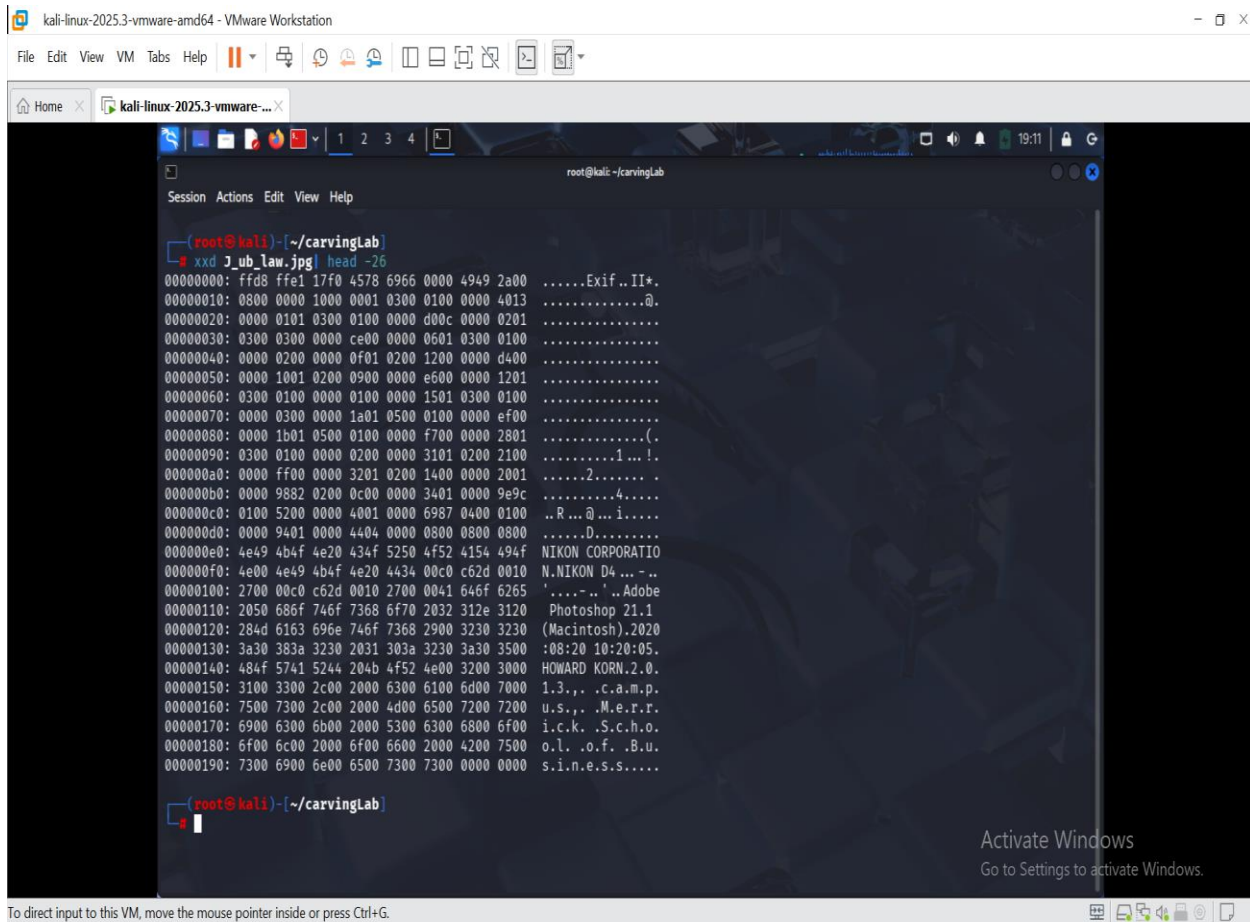
The terminal window includes a menu bar (Session, Actions, Edit, View, Help) and a taskbar at the bottom with various application icons. An 'Activate Windows' watermark is visible in the bottom right corner.

Using the command: `xxd -p J_ub_law.jpg | tr -d "\n" | more`



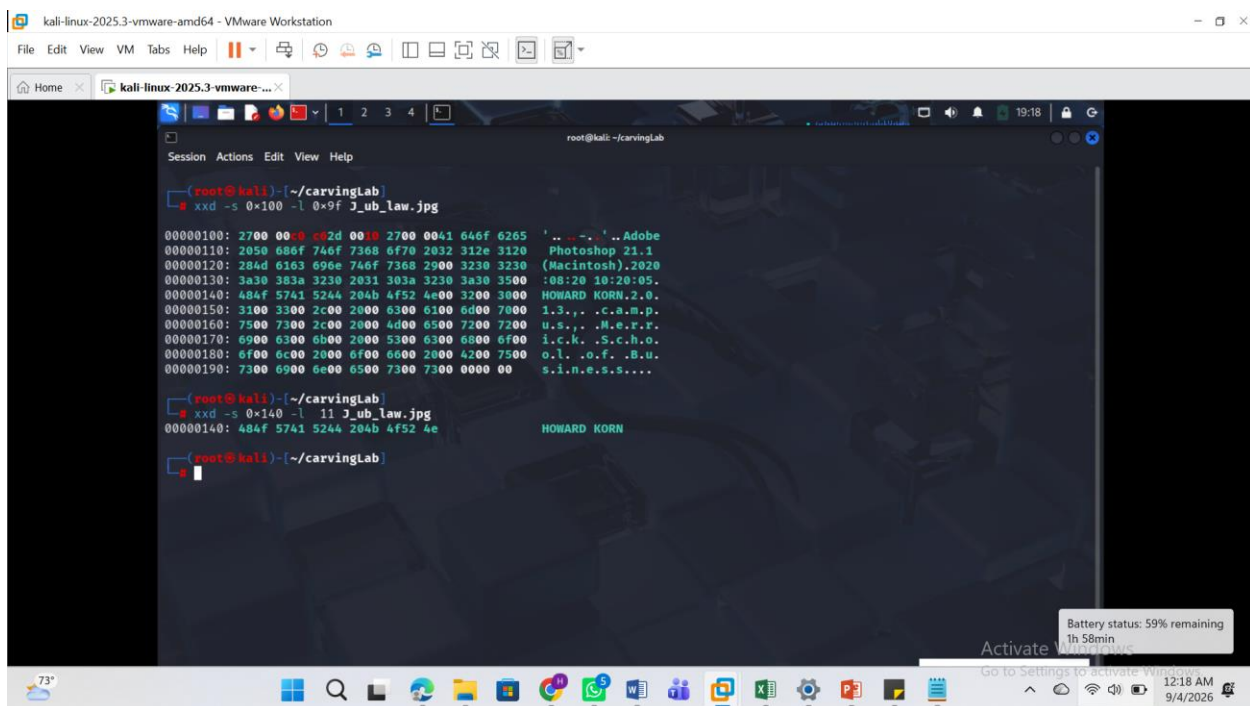
## Convert a plain hexdump to back into binary

**See the command used:** xxd -r -p J\_ub\_law\_hexdump.txt  
J\_ub\_law\_reverse\_dum.jpg



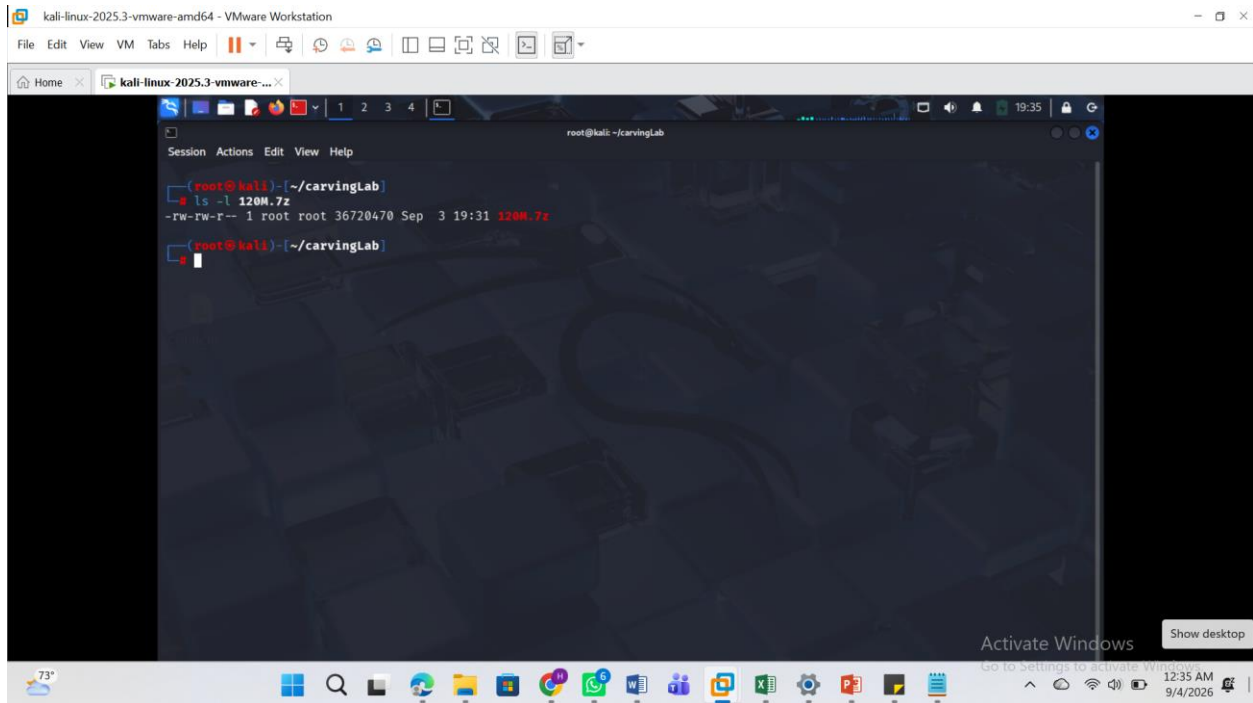
**xxd -s 0x100 -l 0x9f J\_ub\_law.jpg**





**Verifying the reconstructed image by comparing MD5 and SHA-256 hashes with the original**

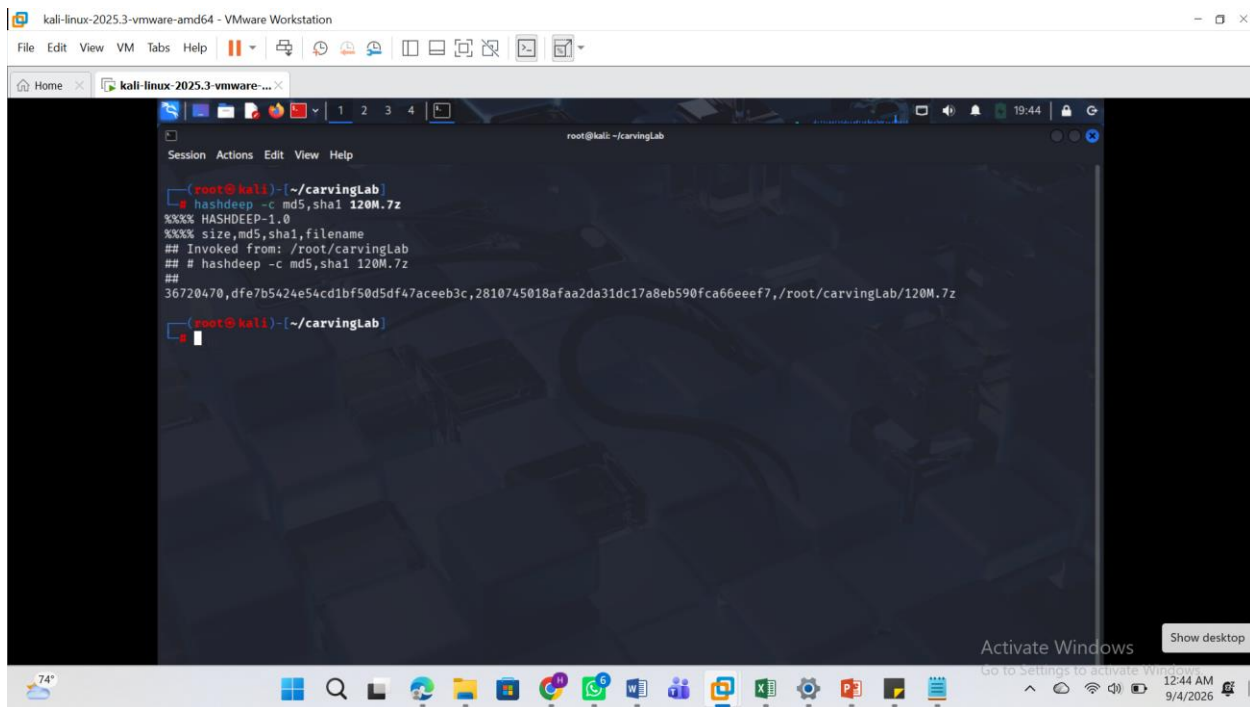
Command used: `ls -l 120M.7z` showing the root kali permissions



**comparing MD5 and SHA-256 hashes with the original**

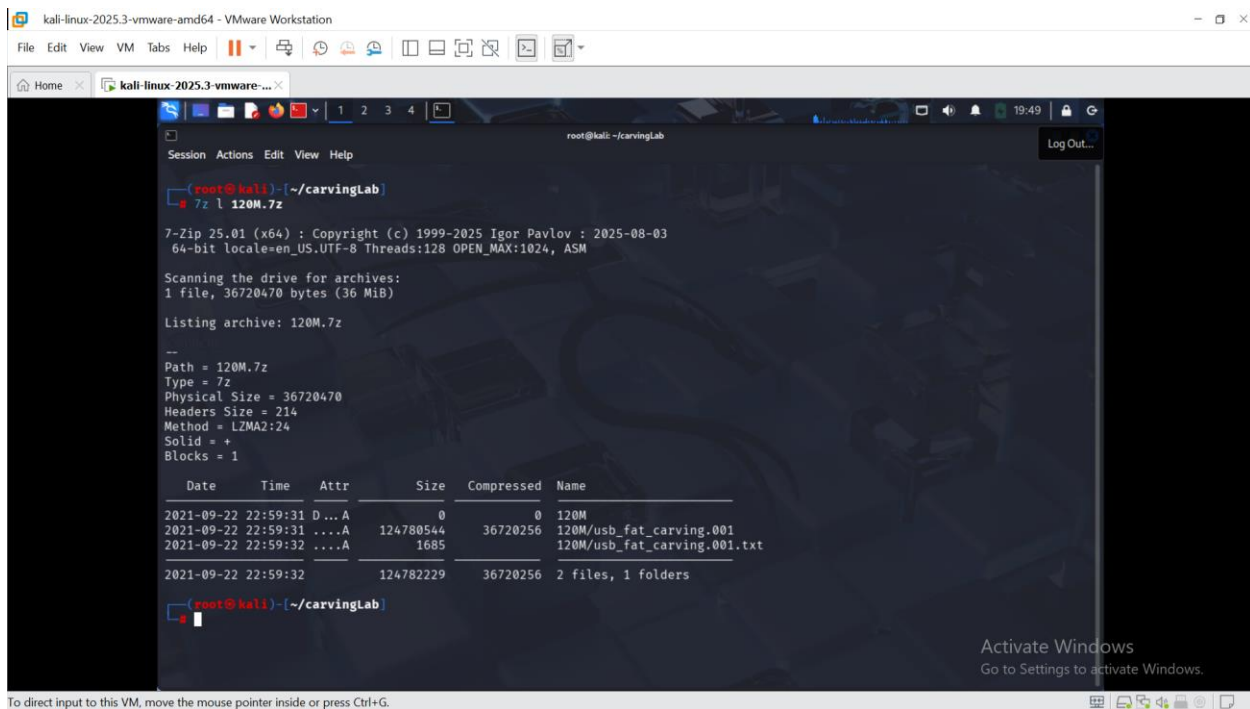
see attach and the command used: `hashdeep -c md5,sha1 120M.7zclear`



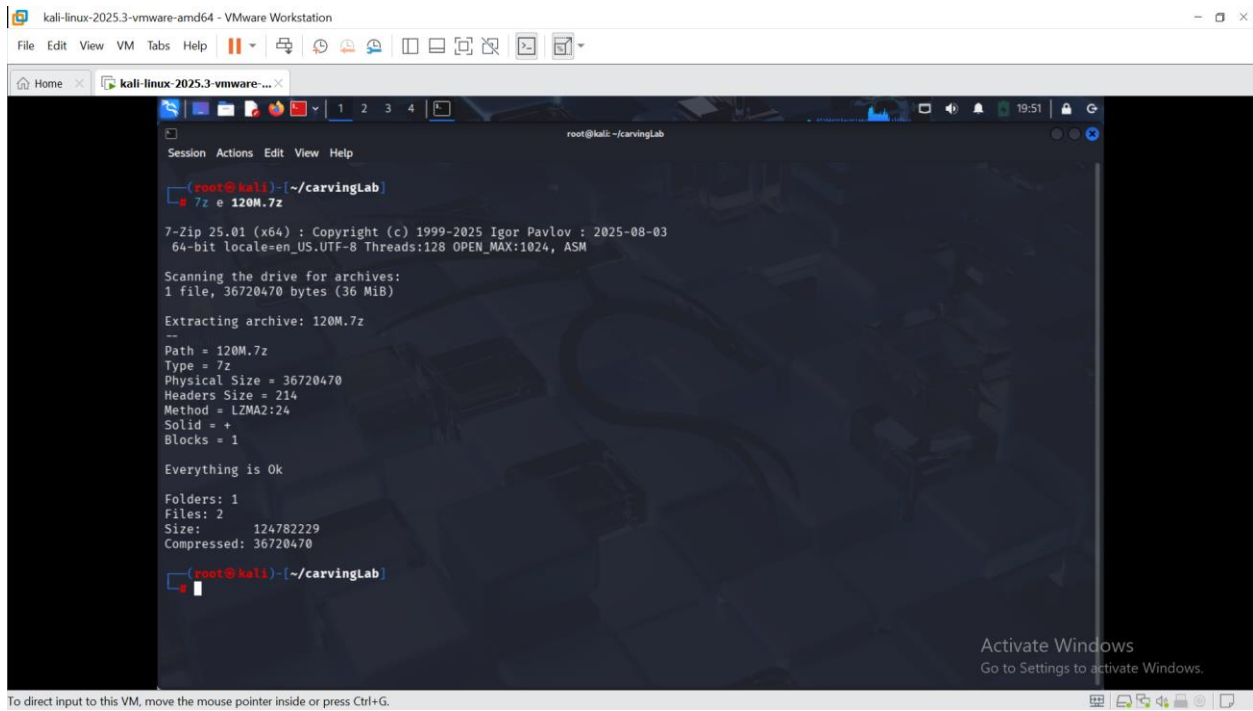


Listing the content of the zip file using the command: 7z l 120M.7z

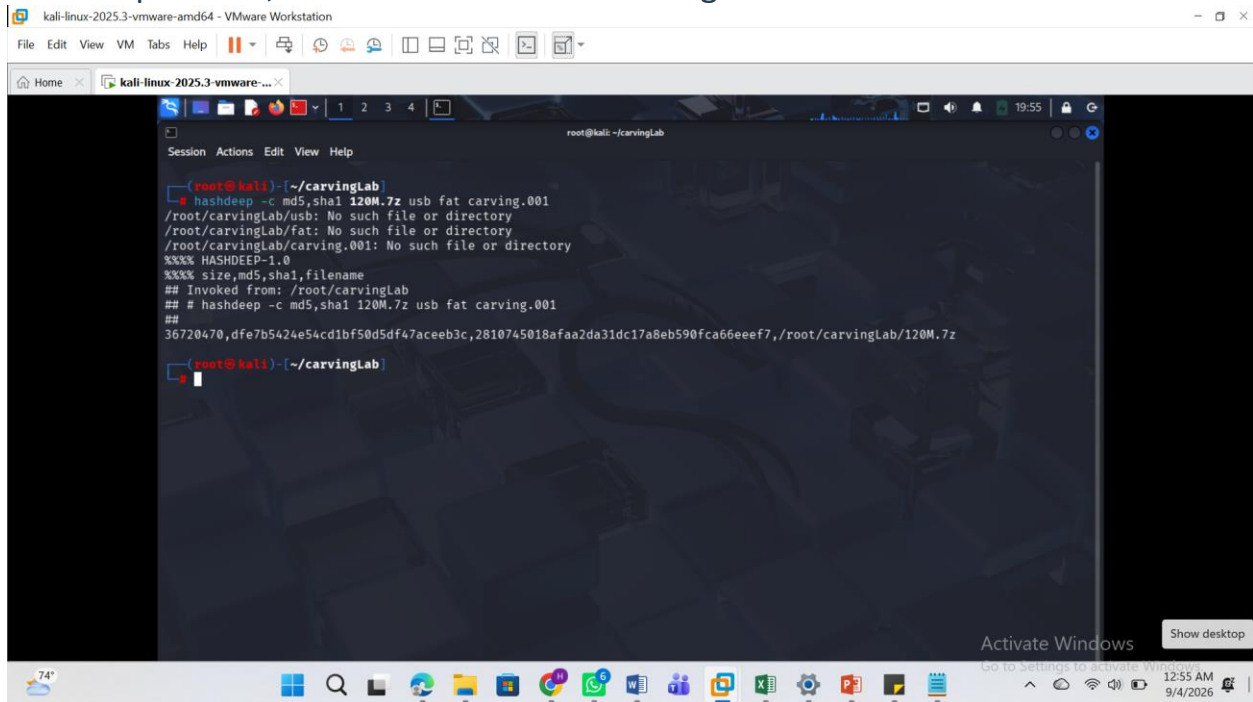
See output below

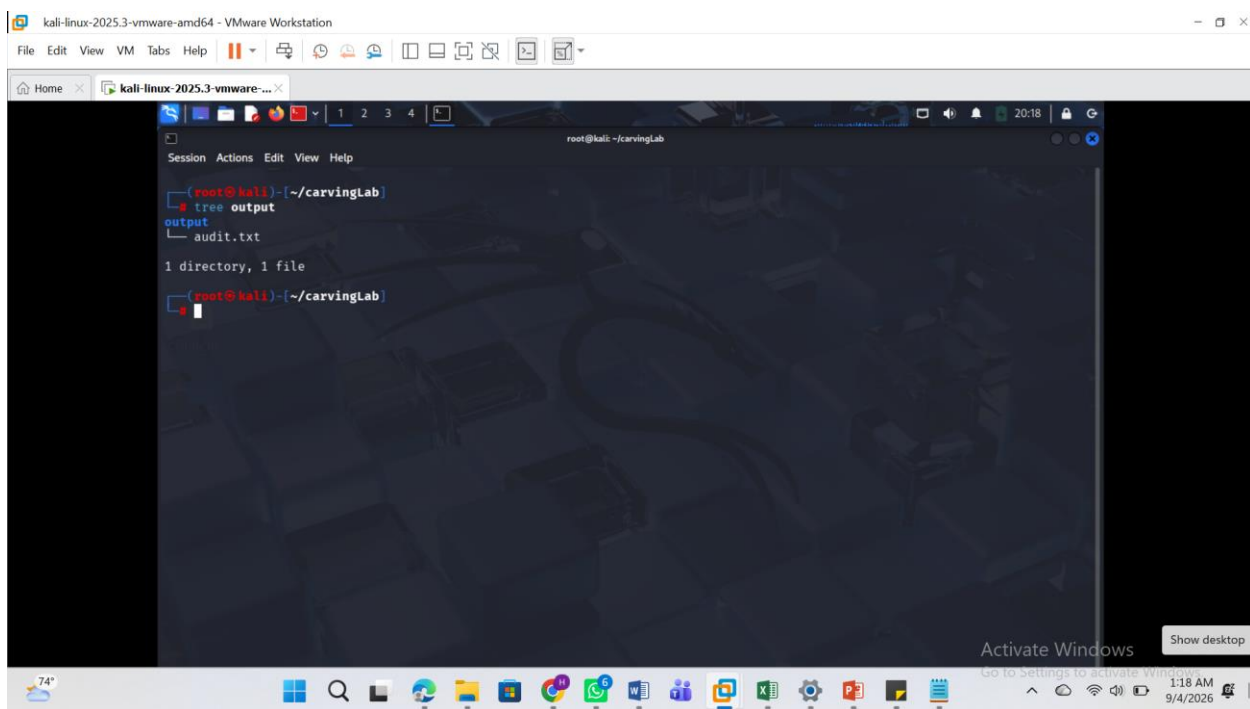
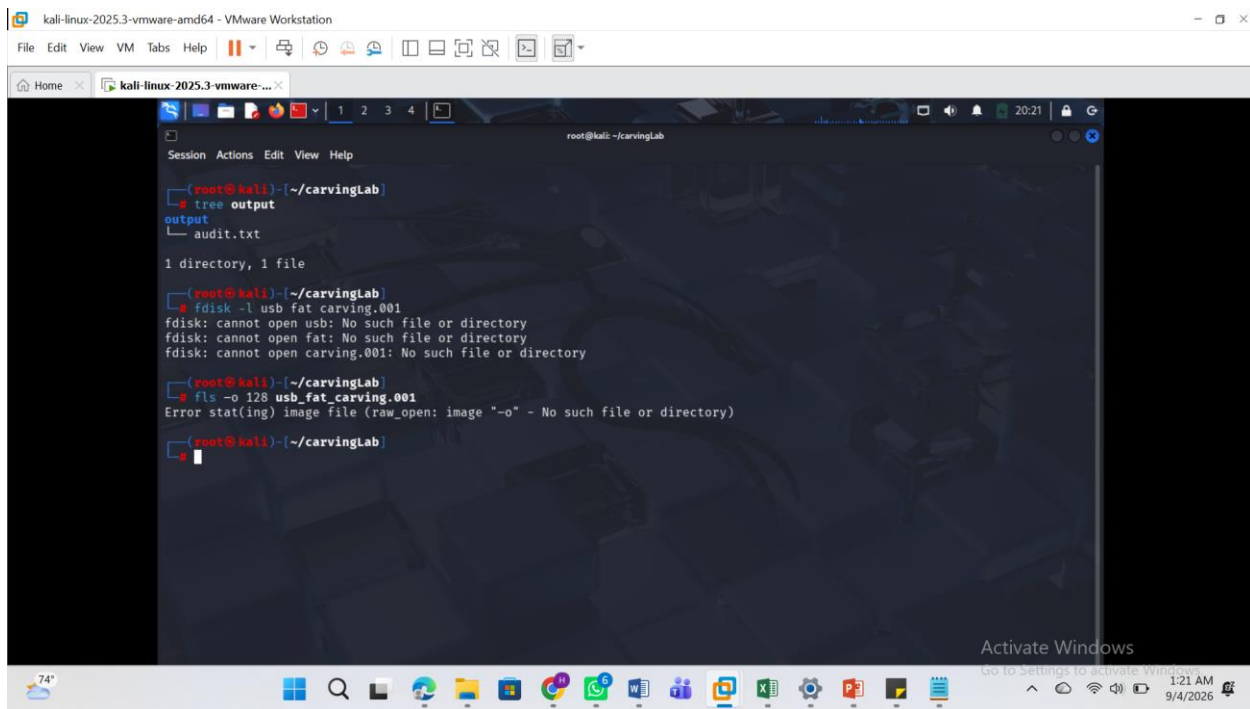


7z e 120M.7z

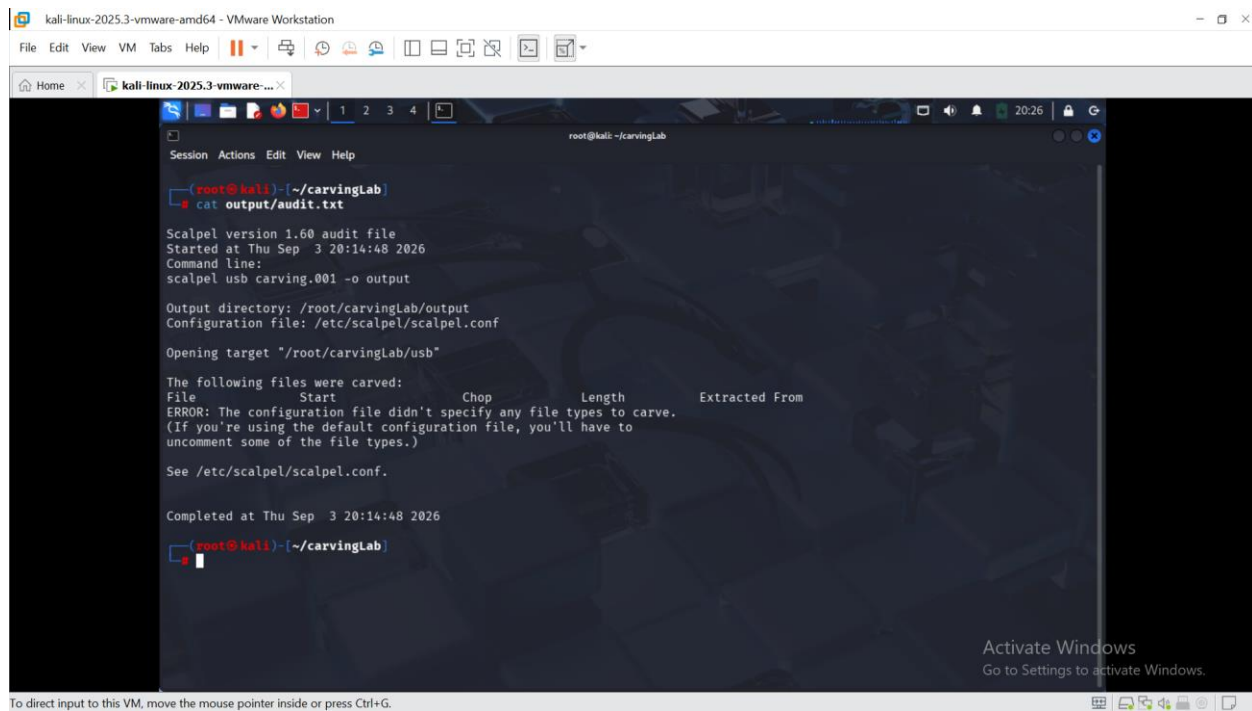


## hashdeep -c md5,sha1 120M.7z usb fat carving.001





cat output/audit.txt



The screenshot shows a Kali Linux terminal window titled "kali-linux-2025.3-vmware-amd64 - VMware Workstation". The terminal is running the Scalpel tool. The user has entered the command `cat output/audit.txt`. The output of the command is as follows:

```
Scalpel version 1.60 audit file
Started at Thu Sep 3 20:14:48 2026
Command line:
scalpel usb carving.001 -o output

Output directory: /root/carvingLab/output
Configuration file: /etc/scalpel/scalpel.conf

Opening target "/root/carvingLab/usb"

The following files were carved:
File          Start          Chop          Length          Extracted From
ERROR: The configuration file didn't specify any file types to carve.
(If you're using the default configuration file, you'll have to
uncomment some of the file types.)

See /etc/scalpel/scalpel.conf.

Completed at Thu Sep 3 20:14:48 2026
```

At the bottom of the terminal window, there is a message: "Activate Windows Go to Settings to activate Windows." Below the terminal window, there is a note: "To direct input to this VM, move the mouse pointer inside or press Ctrl+G."

## Part B - Examine Embedded Metadata-Like Content

Use `xxd` and `strings` to inspect the JPEG for visible camera, software, name, date or other metadata-related strings.

Record only information actually found. If a searched item is not present, state **“not found”** rather than guessing.

## Part C - Direct Data-Unit Examination with The Sleuth Kit

Use tools such as:

- `img_stat`
- `mmls`
- `fsstat`
- `fls`
- `icat`
- `blkcat`

Identify actual inode and sector/block values from your own command output. Do not copy placeholder values from examples.

### **Part D - Embedded File Discovery with Binwalk**

Use binwalk to identify embedded file structures or signatures.

If the instructor-provided File\_carving.docx is available, perform the demonstrated manual and automatic extraction workflow. If it is not available, document this limitation and complete the Binwalk practice using the other supplied lab evidence.

### **Part E - Prepare the USB Image**

Extract 120M.7z, identify the resulting USB image, calculate hashes, inspect the image using TSK, and confirm the appropriate partition offset using mmls.

The source lab refers to an offset of 128, but students must confirm the partition structure rather than blindly copying the value.

### **Part F - Carve Deleted Files with Scalpel**

Configure Scalpel to enable the required file types, particularly JPEG/JPG, and run the carving operation against the forensic USB image.

Review:

- Recovered files
- File types
- Hashes
- audit.txt
- At least two recovered JPEG images where available

Document every command, result, screenshot, recovered artefact, error and limitation.

**Do not modify the original evidence files.**

**Required evidence**

Submit one professional PDF forensic report and one ZIP archive containing the required recovered/carved evidence files, hash records and supporting output.

Required evidence must include:

Proof that the Module 7 material was reviewed before starting.

Working directory and downloaded evidence files with file sizes and hashes.

xxd output showing JPEG header and footer analysis.

Reverse-hexdump reconstruction and hash comparison.

strings or metadata-style examination of the sample JPEG.

TSK mmls / fls and related data-unit analysis output.

Scalpel configuration, execution output and audit.txt.

At least two recovered JPEG files opened successfully, or clear evidence explaining why they could not be opened.

**Lab access notes** This practical forms part of the second laboratory assessment window. Students must review the Module 7 lessons and study material before beginning. Use only the authorised training files provided in the lab instructions. Preserve the original evidence, record hashes, document all commands and submit both the required PDF report and supporting evidence archive.